Applied Machine Learning and Predictive Modelling 1

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Model Validation

How to select the "best" model?

- In modelling we have to make many choices (which model? which predictors? non-linear effects? interactions? etc.)
- For prediction, we want a model not to overfit our dataset such that it performs well on new/future data.
- Models should be complex enough, but not too much to overfit to the training data.
- We do not have access to new data, just have the dataset at hand.
- Thus, we can only **estimate** the generalization performance of a model on new data.
- Based on this estimate we can then select the "best" model.

Cross-Validation

See the "Cross Validation Lab" and corresponding exercises (not exam material).

How to select the "best" model?

In very many settings the **predictive performance of a model is blindly taken as the only thing to maximise**...
Data Science has much more to offer!

Among the many candidate models, the "best" one is the one that best fulfils the aims of the project and the actual uses of the model and ...
not the one that simply maximises e.g. RMSE/AUC in 10-fold CV!

Ask yourself: How will the model be used?

Elements to consider when selecting the “best” model

Example 1: Amazon recommender system

- Once you click on an item, there are several proposals of other related items you might be interested in.
- The proposals come from a model called “recommender system”.
- What are the properties that such a model should have?

Elements to consider when selecting the “best” model

Example 2: Predictions for patients in another hospital

- Doctors want to decide if a patient should or should not undergo a surgery for pelvis fracture.
- The hospital fitted a model with 70 predictors such as “age”, “gender”, “type of fracture”, “MRI images”, “diagnosis”, and validated it in-sample.
- The hospital claims the model can be used in other hospitals, too.
- Can the model be used in other hospitals?

Elements to consider when selecting the “best” model

Example 3: Predictions using weather as predictor

- A company wants to predict how many clients will walk in their shops in two weeks time (to allocate their staff).
- They use “week of the year [1-52]”, “day of the week [Mo-Sa]”, “holiday [yes,no]”, “weather [cloudy, sunny, ...]” and a few more predictors.
- Is this model feasible in practice?

Elements to consider when selecting the "best" model

Example 4: Predictions for new geographic regions

- We want to predict costs of public transportation companies in Swiss cantons.
- The costs will be paid by subventions (money paid from the Swiss governement).
- Concretely, the aim is to estimate the costs of a bus line, based on "line length", "bus type used", "bus frequency", "road average steepness", "electric or fuel" etc.
- A model is trained with data from ZH, BE, AG, SG, and TI.
- Next year, GE joins the project and the cost of all its lines need to be estimated based on the model (trained on ZH, BE, AG, SG, and TI).
- So we want to extrapolate the model's predictions to GE.
- How would you validate the predictive performance of our model?

Elements to consider when selecting the “best” model

Example 5: Trust of regulators in FINMA

- We want to train a predictive model for FINMA (Swiss Financial Market Supervisory Authority).
- We observed that neural nets performed best.
- What has to be considered when proposing this model?

Elements to consider when selecting the "best" model

Example 6: Can you find an example from your own work experience?

Long-term THM

- **The validation should simulate the setting in which the model is used.**
- **Make sure you understand the constraints of how the model is used in practice.**
- The choice of the most suited method depends on the situation.
- Once you have considered all practical elements, then only "maximizing performance" becomes relevant...
- Often several methods are used simultaneously (to compare results).
- **The best model is the most useful one!**